MAHARASHTRA EDUCATION SOCIETY INSTITUTE OF MANAGEMENT AND CAREER COURSES

MES IMCC, 131, Mayur Colony, Kothrud,Pune

College Code: INST-435 • Pune • Maharashtra

VALUE ADDED DSA

Official Academic Assessment Evaluation Transcript • Class: Division A (A)

STATEMENT OF MARKS & PERFORMANCE REPORT — Roll No: 2601103 | Exam Date: 23 Oct 2026

Tanvi Khedekar

Roll Number: 2601103 **Email:** tanvi.jkhedekar1@gmail.com

Classroom: Division A (A) **Duration:** 40 minutes **Set:** Set A

Status: COMPLETED & OFFICIALLY EVALUATED

OFFICIAL COMPOSITE SCORE

31 / 40

Aggregate: 78%

#	Question / Problem & Candidate Response	Max Marks	Marks Awarded	Status / Remarks
Q1	If head is NULL and traversal uses while(temp != NULL), what happens? A. The loop runs forever B. The loop body does not execute <- Candidate's choice [Correct] C. A node is inserted D. A node is deleted Candidate Answer: The loop body does not execute | Correct Answer: The loop body does not execute	1	1	Correct [OK]
Q2	For list 4 -> 8 -> 12 -> NULL, print values divisible by 4. What is printed? A. 4 8 12 <- Candidate's choice [Correct] B. 8 12 C. 4 12 D. Nothing Candidate Answer: 4 8 12 | Correct Answer: 4 8 12	1	1	Correct [OK]
Q3	What is printed by this Python code? stack = [3, 6, 9]; print(stack.pop(), stack.pop(), stack[-1]) A. 3 6 9 B. 9 6 3 <- Candidate's choice [Correct] C. 9 3 6 D. 6 9 3 Candidate Answer: 9 6 3 | Correct Answer: 9 6 3	1	1	Correct [OK]
Q4	What is printed by this Python code? q = [4, 8, 12]; front = 1; rear = 2; print(q[front], q[rear]) A. 4 8 B. 8 12 <- Candidate's choice [Correct] C. 4 12 D. 12 8 Candidate Answer: 8 12 | Correct Answer: 8 12	1	1	Correct [OK]
Q5	What should happen if pop() is called on an empty stack? A. Remove arbitrary memory B. Handle stack underflow <- Candidate's choice [Correct] C. Push zero automatically D. Set capacity to zero Candidate Answer: Handle stack underflow | Correct Answer: Handle stack underflow	1	1	Correct [OK]
Q6	Which operation removes the top element from a stack? A. push() B. pop() <- Candidate's choice [Correct] C. front() D. enqueue() Candidate Answer: pop() | Correct Answer: pop()	1	1	Correct [OK]
Q7	What is the time complexity of pop on a linked-list stack with top at head? A. O(n) <- Candidate's choice B. O(n^2) C. O(1) [Correct] D. O(log n) Candidate Answer: O(n) | Correct Answer: O(1)	1	0	Incorrect
Q8	For an array-based stack using top=-1 initially, after one successful push, what is top? A. -1 B. 0 <- Candidate's choice [Correct] C. 1 D. 5 Candidate Answer: 0 | Correct Answer: 0	1	1	Correct [OK]
Q9	What is printed by this Python code? q = [1, 2, 3]; rear = 2; print(q[rear]) A. 1 B. 2 C. 3 <- Candidate's choice [Correct] D. 0 Candidate Answer: 3 | Correct Answer: 3	1	1	Correct [OK]
Q10	Why are head insertion and deletion suitable for a linked-list stack? A. Both can be O(1) <- Candidate's choice [Correct] B. They sort values C. They prevent allocation D. They allow indexed access in O(1) Candidate Answer: Both can be O(1) | Correct Answer: Both can be O(1)	1	1	Correct [OK]

#	Question / Problem & Candidate Response	Max Marks	Marks Awarded	Status / Remarks
Q11	Appending to a singly linked list without a tail pointer generally takes what time for n nodes? A. O(1) B. O(log n) C. O(n) [Correct] D. O(n^2) Candidate Answer: Not Answered | Correct Answer: O(n)	1	0	Unattempted
Q12	The list is 10 -> 20 -> NULL. Insert 30 at the end. What is the result? A. 30 -> 10 -> 20 -> NULL B. 10 -> 30 -> 20 -> NULL C. 10 -> 20 -> 30 -> NULL <- Candidate's choice [Correct] D. 10 -> 20 -> NULL Candidate Answer: 10 -> 20 -> 30 -> NULL | Correct Answer: 10 -> 20 -> 30 -> NULL	1	1	Correct [OK]
Q13	Which sequence removes all elements from a stack containing 10 bottom, 20 middle, 30 top? A. pop(), pop(), pop() [Correct] B. push(10), push(20), push(30) C. peek(), peek(), peek() D. enqueue(), dequeue(), dequeue() Candidate Answer: Not Answered | Correct Answer: pop(), pop(), pop()	1	0	Unattempted
Q14	What is printed by this Python code? stack = []; stack.append(15); stack.append(25); print(stack[-1]) A. 15 B. 25 [Correct] C. 0 D. -1 Candidate Answer: Not Answered | Correct Answer: 25	1	0	Unattempted
Q15	In a linear array queue, what does front=rear=-1 initially commonly indicate? A. Full queue B. Empty queue <- Candidate's choice [Correct] C. One item D. Wrapped-around queue Candidate Answer: Empty queue | Correct Answer: Empty queue	1	1	Correct [OK]
Q16	An array stack of capacity 4 starts empty. After three pushes, what is top? A. -1 B. 0 <- Candidate's choice C. 2 [Correct] D. 4 Candidate Answer: 0 | Correct Answer: 2	1	0	Incorrect
Q17	Stack starts empty: push(4), push(8), pop(), push(6), peek(). What is returned? A. 4 B. 6 <- Candidate's choice [Correct] C. 8 D. 0 Candidate Answer: 6 | Correct Answer: 6	1	1	Correct [OK]
Q18	Which two pointers are commonly maintained in a linked-list queue? A. head and middle B. front and rear <- Candidate's choice [Correct] C. top and bottom D. left and right Candidate Answer: front and rear | Correct Answer: front and rear	1	1	Correct [OK]
Q19	Queue starts empty: enqueue(10), enqueue(20), dequeue(), dequeue(). What is its state? A. Contains 10 B. Contains 20 C. Empty <- Candidate's choice [Correct] D. Contains both Candidate Answer: Empty | Correct Answer: Empty	1	1	Correct [OK]
Q20	In the array-based stack convention used here, what does top = -1 commonly indicate? A. One element exists B. The stack is full C. The stack is empty <- Candidate's choice [Correct] D. Capacity is negative Candidate Answer: The stack is empty | Correct Answer: The stack is empty	1	1	Correct [OK]
Q21	Which sequence makes a newly created node the first node of an empty linked list? A. head->next = newNode; <- Candidate's choice B. newNode->next = head; head = newNode; [Correct] C. newNode = head->next; D. head = head->next; Candidate Answer: head->next = newNode; | Correct Answer: newNode->next = head; head = newNode;	1	0	Incorrect
Q22	Which loop reaches the last node of a non-empty singly linked list? A. while(temp != NULL) temp=temp->next; then use temp->next B. while(temp->next != NULL) temp=temp->next; <- Candidate's choice [Correct] C. while(head == NULL) temp=temp->next; D. temp=NULL; Candidate Answer: while(temp->next != NULL) temp=temp->next; | Correct Answer: while(temp->next != NULL) temp=temp->next;	1	1	Correct [OK]
Q23	What must be handled separately when deleting the first node? A. The list is empty <- Candidate's choice [Correct] B. The list has a thousand nodes only C. All values are different D. The last value is even Candidate Answer: The list is empty | Correct Answer: The list is empty	1	1	Correct [OK]
Q24	A linked-list stack is 30 -> 20 -> 10, with 30 at top. After one pop, what remains? A. 30 -> 20 -> 10 B. 20 -> 10 <- Candidate's choice [Correct] C. 10 -> 20 D. NULL Candidate Answer: 20 -> 10 | Correct Answer: 20 -> 10	1	1	Correct [OK]

#	Question / Problem & Candidate Response	Max Marks	Marks Awarded	Status / Remarks
Q25	Which operation adds an element to a queue? A. push() B. pop() C. enqueue() <- Candidate's choice [Correct] D. peekStack() Candidate Answer: enqueue() | Correct Answer: enqueue()	1	**1**	Correct [OK]
Q26	A fixed-capacity linear queue is full. What should another enqueue do if no capacity is available? A. Handle overflow <- Candidate's choice [Correct] B. Always remove oldest automatically C. Become a stack D. Always reset front to zero Candidate Answer: Handle overflow | Correct Answer: Handle overflow	1	**1**	Correct [OK]
Q27	Insert 20 after 10 in 10 -> 30 -> 40 -> NULL. What is the result? A. 20 -> 10 -> 30 -> 40 -> NULL B. 10 -> 20 -> 30 -> 40 -> NULL <- Candidate's choice [Correct] C. 10 -> 30 -> 20 -> 40 -> NULL D. 10 -> 30 -> 40 -> 20 -> NULL Candidate Answer: 10 -> 20 -> 30 -> 40 -> NULL | Correct Answer: 10 -> 20 -> 30 -> 40 -> NULL	1	**1**	Correct [OK]
Q28	For capacity MAX and zero-based indexing, when is an array-based stack full? A. top == -1 B. top == 0 C. top == MAX - 1 <- Candidate's choice [Correct] D. top == MAX + 1 Candidate Answer: top == MAX - 1 | Correct Answer: top == MAX - 1	1	**1**	Correct [OK]
Q29	What is printed by this Python code? q = [10, 20]; front = 0; print(q[front]) A. 10 <- Candidate's choice [Correct] B. 20 C. 0 D. -1 Candidate Answer: 10 | Correct Answer: 10	1	**1**	Correct [OK]
Q30	Why maintain both front and rear pointers in a linked-list queue? A. To sort all values B. Efficient insertion at rear and deletion at front <- Candidate's choice [Correct] C. To behave like a stack D. To prevent becoming empty Candidate Answer: Efficient insertion at rear and deletion at front | Correct Answer: Efficient insertion at rear and deletion at front	1	**1**	Correct [OK]
Q31	A stack is represented by a Python list [5, 10]. After one pop, how many elements remain? A. -1 B. 0 [Correct] C. 1 <- Candidate's choice D. 2 Candidate Answer: 1 | Correct Answer: 0	1	**0**	Incorrect
Q32	A list has exactly one node. After correctly deleting it, what should head contain? A. Deleted node's address B. Address of a new node C. NULL <- Candidate's choice [Correct] D. Deleted node's value Candidate Answer: NULL | Correct Answer: NULL	1	**1**	Correct [OK]
Q33	An empty queue receives enqueue(5) then dequeue(). What is its state? A. Contains 5 B. Empty <- Candidate's choice [Correct] C. Contains -1 D. Contains two elements Candidate Answer: Empty | Correct Answer: Empty	1	**1**	Correct [OK]
Q34	A queue contains 5, 8, 12 with 5 at front. After one dequeue, what is front? A. 5 B. 8 <- Candidate's choice [Correct] C. 12 D. Empty Candidate Answer: 8 | Correct Answer: 8	1	**1**	Correct [OK]
Q35	head is 10 -> 20 -> 30 -> NULL. After temp=head; temp=temp->next; temp=temp->next;, what is temp->data? A. 10 B. 20 C. 30 <- Candidate's choice [Correct] D. NULL Candidate Answer: 30 | Correct Answer: 30	1	**1**	Correct [OK]
Q36	Which principle does a stack follow? A. FIFO B. LIFO <- Candidate's choice [Correct] C. Random access only D. Sorted order Candidate Answer: LIFO | Correct Answer: LIFO	1	**1**	Correct [OK]
Q37	An array stack starts top=-1. After push(4), push(7), push(9), what is top? A. -1 B. 0 C. 2 <- Candidate's choice [Correct] D. 3 Candidate Answer: 2 | Correct Answer: 2	1	**1**	Correct [OK]
Q38	For list 10 -> 20 -> 30 -> NULL, what does printing each node's data multiplied by 2 produce? A. 10 20 30 B. 20 40 60 <- Candidate's choice [Correct] C. 12 22 32 D. 60 40 20 Candidate Answer: 20 40 60 | Correct Answer: 20 40 60	1	**1**	Correct [OK]

#	Question / Problem & Candidate Response	Max Marks	Marks Awarded	Status / Remarks
Q39	What is printed by this Python code? stack = [11, 22]; print(stack[-1], stack[-1]) A. 11 22 B. 22 11 C. 22 22 [Correct] D. 11 11 Candidate Answer: Not Answered | Correct Answer: 22 22	1	**0**	Unattempted
Q40	What is printed by this Python code? q = [10, 20, 30]; front = 1; print(q[front]) A. 10 B. 20 [Correct] C. 30 D. 0 Candidate Answer: Not Answered | Correct Answer: 20	1	**0**	Unattempted
	TOTAL COMPOSITE MARKS	**40**	**31 / 40**	**Percentage: 78%**

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